Project Report V2

F-16 Fighting Falcon

Aerodynamics 1

AERO 3312

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Airfoil

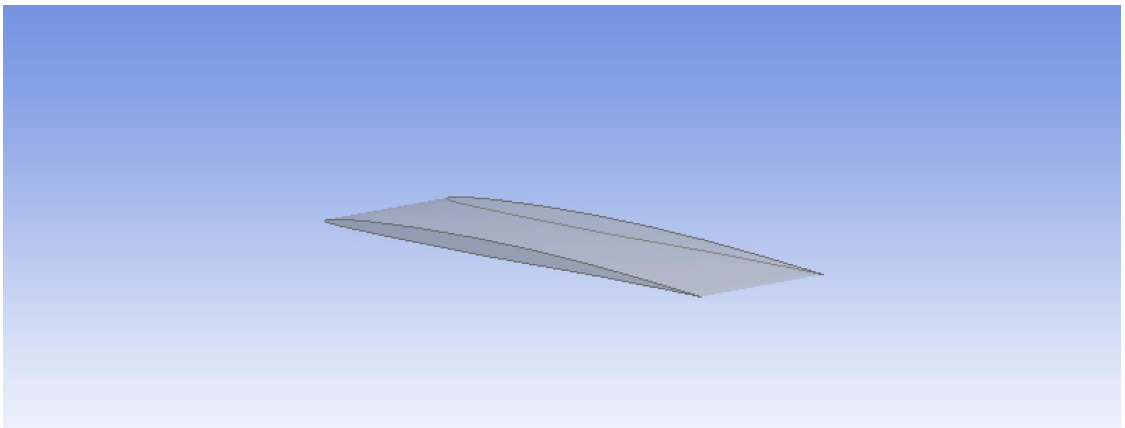
Introduction

The F-16 Fighting Falcon is a supersonic aircraft originally designed by *General Dynamics* and now produced by *Lockheed Martin* for the United States Air Force as well as other countries. While having a body length of 15 meters and a wingspan of 9.45 meters, the F-16 is powered by a *Pratt & Whitney* or *General Electric* turbo fan engine capable of generating 130 kilonewtons of thrust and pushing the aircraft to a speed of Mach 2. Under a typical combat load, the aircraft weighs approximately 10,000 kilograms. The F-16's maximum normal operating altitude is roughly 15 kilometers, but in a special maneuver, it is capable of exceeding 21 kilometers before losing altitude due to lack of power and lift.

At this time, we will be analyzing lift and drag as well as pressure, temperature and velocity plots of the craft's airfoil. These parameters will be observed under 3 scenarios: cruising speed, Mach 1 and Mach 2.

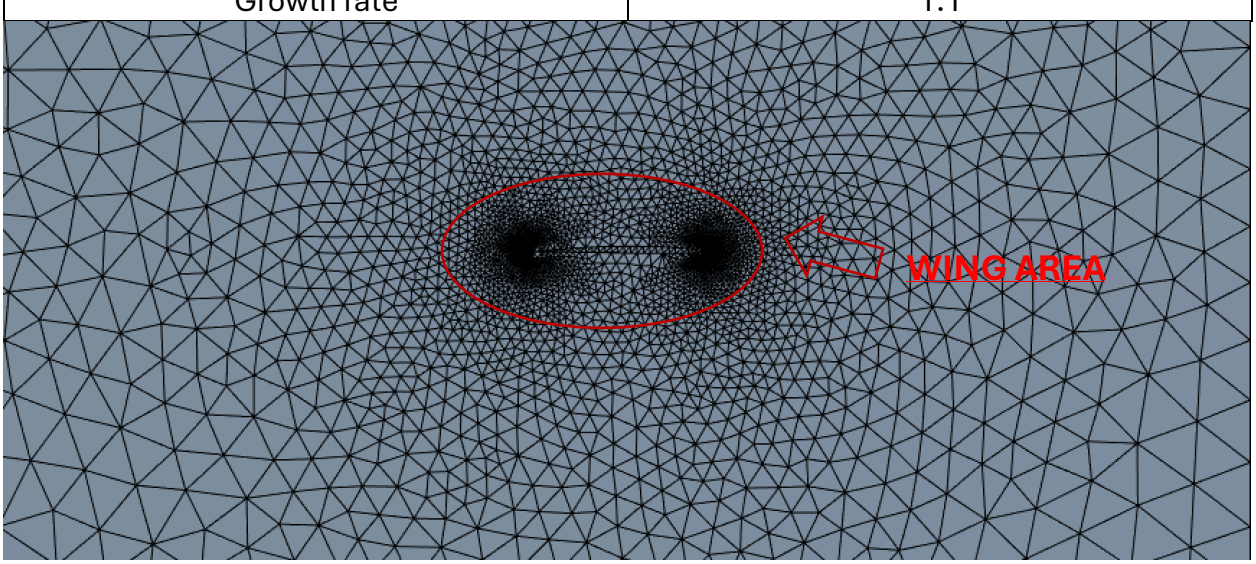
Methodology

For our aircraft, the chosen airfoil model is NACA 64A204. The NACA 64A204 model is a thin, laminar-flow airfoil designed for high-speed aircraft. It has a 4% thickness-to-chord ratio and is optimized for a design lift coefficient of 0.2, with its minimum pressure point located at 40% of the chord. We imported the geometry to Ansys and scaled the body to a realistic airfoil size. The airfoil in the calculations has a chord length equal to the average chord length of the F-16 which is 1.3 meters. As part of the geometry, we also added an enclosure to which we used the Boolean subtraction function so it would facilitate the calculations.



To obtain the mesh, we selected the enclosure and made some edits to the regular mesh. First, we made the element size 0.5m, the growth rate 1.1, and the max size 0.5m. Next, we inserted a face sizing and selected the faces of the wing in the body, turning “capture curvature” on in the process. This refined the mesh around the wing area so the results will be more accurate.

Variable	Selection
Element type	Default(unchanged)
Element size	0.5m
Number of Elements	956648
Mesh Type	3D
Growth rate	1.1



Setting up the problem was probably the trickiest portion of the CFD. After various trials and errors, we managed to find a working formula. First, we use double precision along with four solver processes as the base. First off, we start off with a density-based solver with steady time. In this CFD we had energy and the viscous (SST K-omega) models turned on. Another change made to the set-up was using ideal-gas properties on the air, with Sutherland's three coefficient equation for viscosity. The boundary condition for the inlet is “velocity-inlet” where we set the components of velocity to those of an aircraft with an angle of attack of 4 degrees. The outlet is set to “pressure-outlet". Initialization used is standard initialization computed from the inlet. It is then run for 1000 iterations for accurate data.

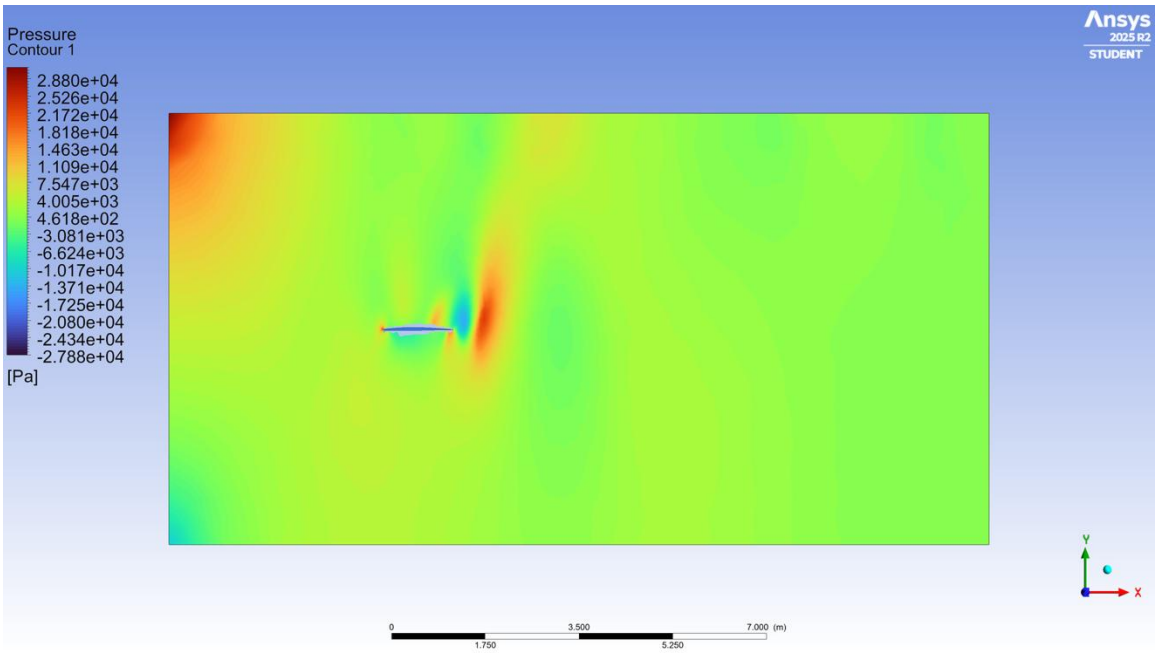
Results

Using both post and fluent we were able to obtain good results. As previously mentioned, we analyzed the pressure, temperature and velocity field, as well as calculated the lift and drag using an Ansys’ built-in calculator. For this first part of the project, we considered three scenarios. These scenarios are cruising, Mach 1, and Mach 2 speeds. Through some research we were able to figure out the cruising speed to be around 275 m/s, and Mach 1 and 2 are 342 m/s and 680 m/s respectively.

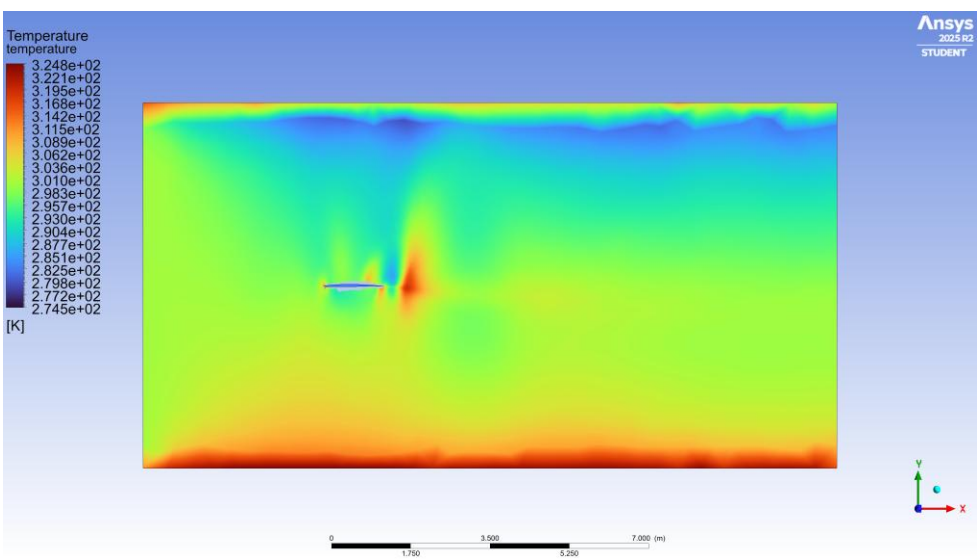
Cruising Speed:

First we will discuss the results from cruising speed. Here we get a drag force of 240[N] and a lift force of 4172[N]. This gives us an L/D ratio of 17.4. The following are the contours:

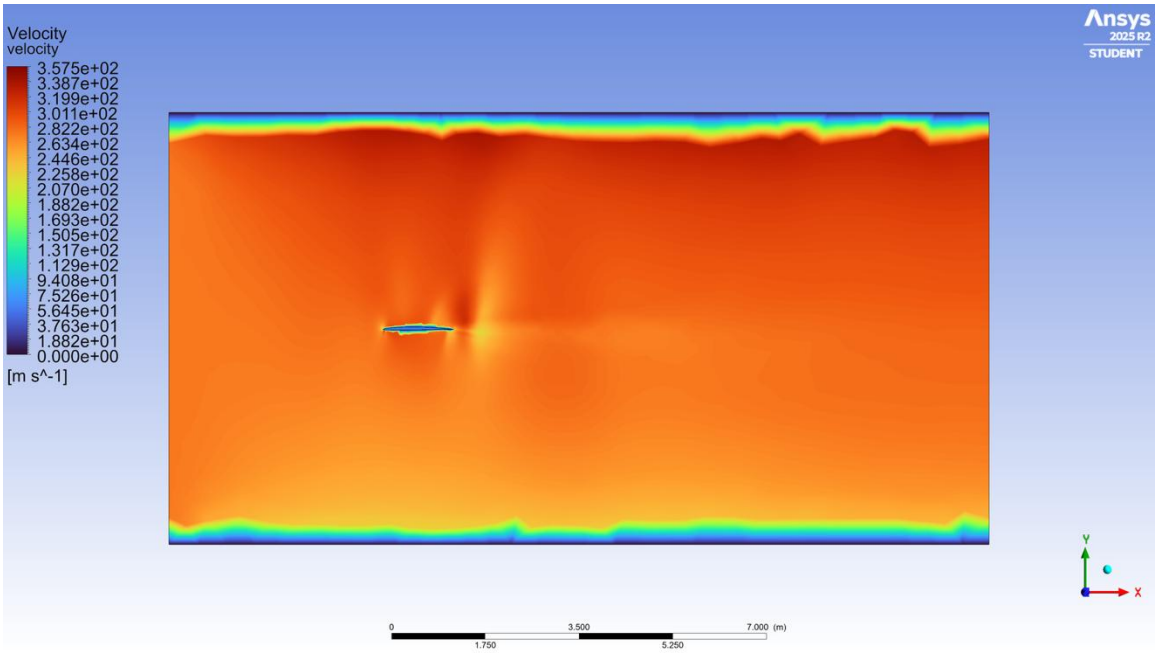
- This would be the pressure contour at a cruising speed. As we can see, we have maximum pressure at the leading edge and minimum pressure at the lower surface of the airfoil.



- This case is the cruise temperature. Like the pressure, it seems to have a max value at the leading edge of the foil and minimum at the lower surface.



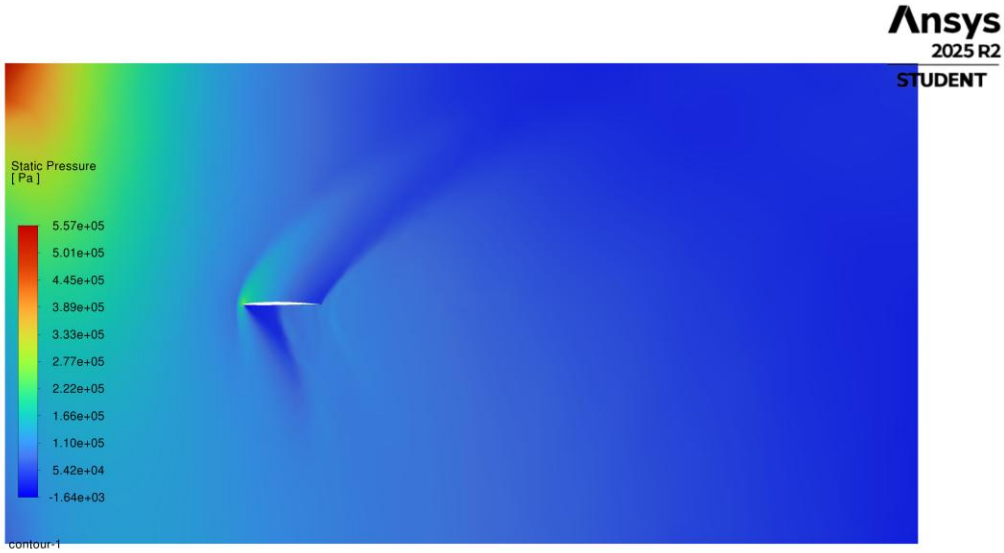
- This is the velocity contour, here we can see how the speeds around the foil look which align with the pressure and temperature contours.



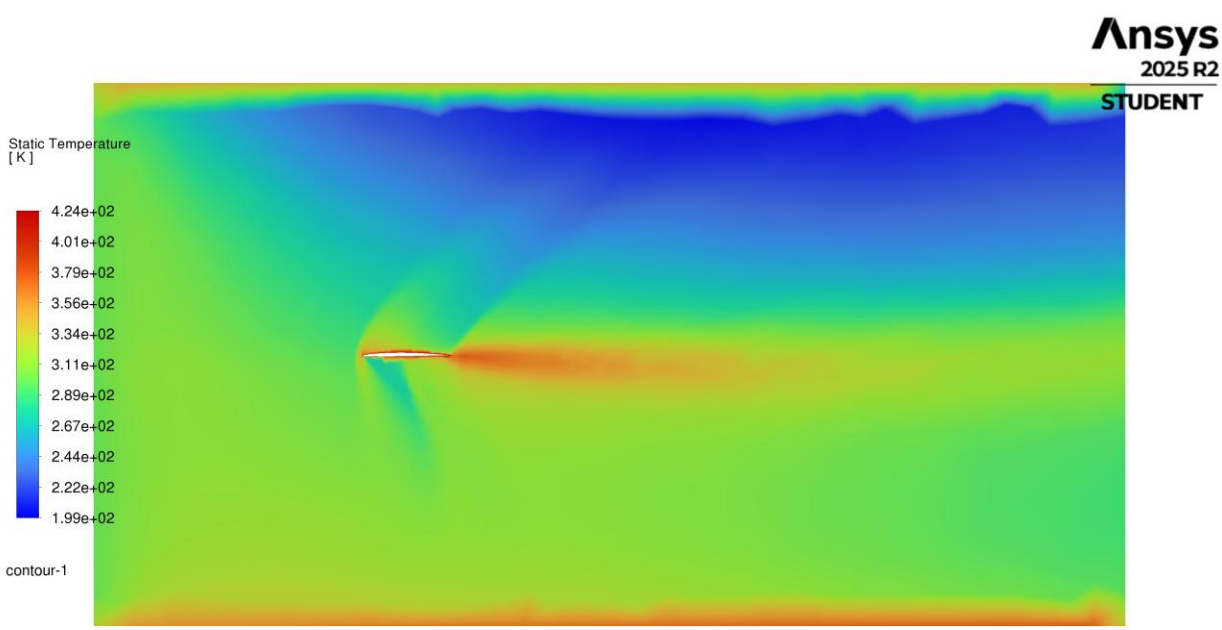
Mach 1:

At Mach 1 we achieved a drag force of 4593[N] and a lifting force of 37175[N]. This gives us an L/D ratio of 8.09. Below will include the pressure, temperature, and velocity contours at this respective speed.

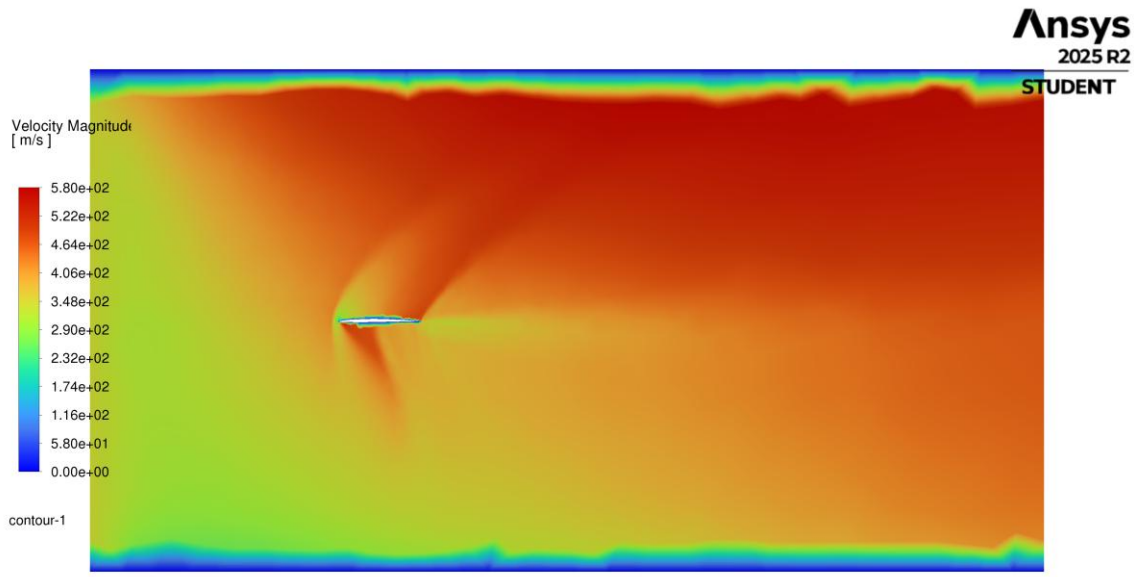
- For static pressure we got a max pressure of around 2.5e5 Pa happening at the leading edge.



- For temperature we have a max temperature happening in the trailing edge of around maybe 3.79e2 kelvin, and a slightly lower temperature of around 3.56e2 Kelvin at the leading edge



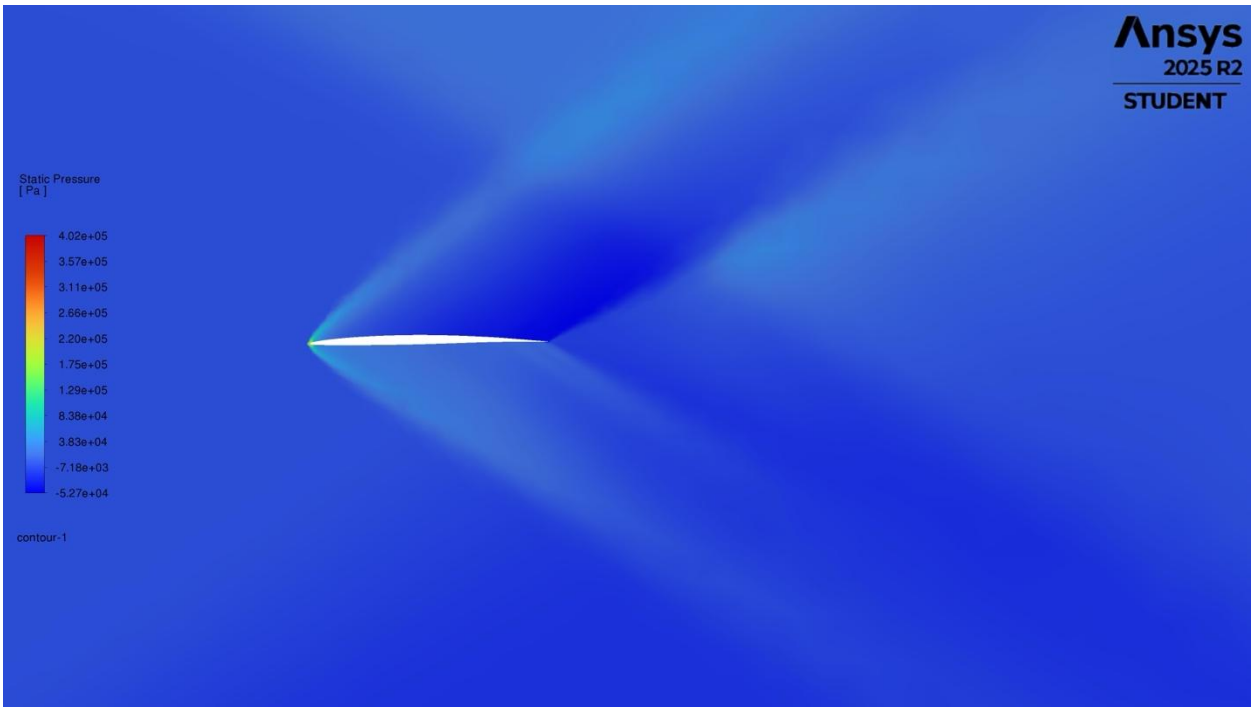
- For velocity we seem to have a lower speed at the leading edge than the rest of the airfoil, looks to be around 2.9e2 m/s



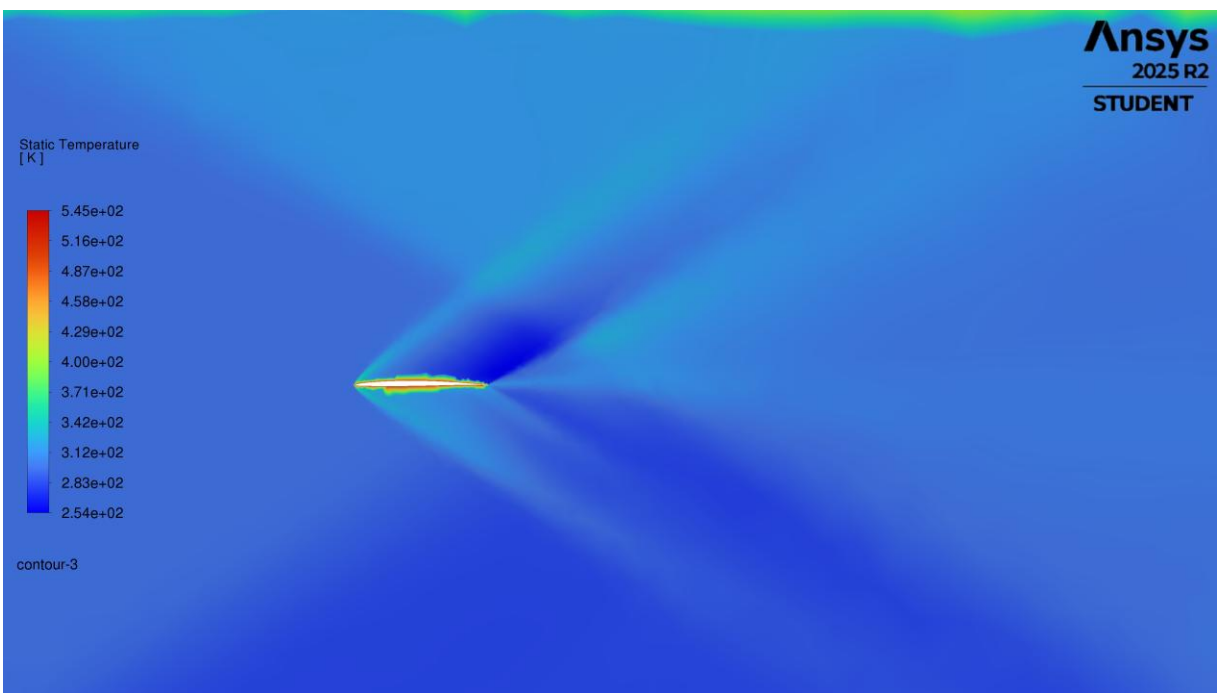
<u>Mach 2:</u>

At Mach 2, calculations gave us a lift of 31529.7 [N] and a 2drag of 4479.04 [N]. Overall, this gave us an L/D ratio of 7.04. This being lower than that of Mach 1. Below are the contours for pressure, temperature, and velocity.

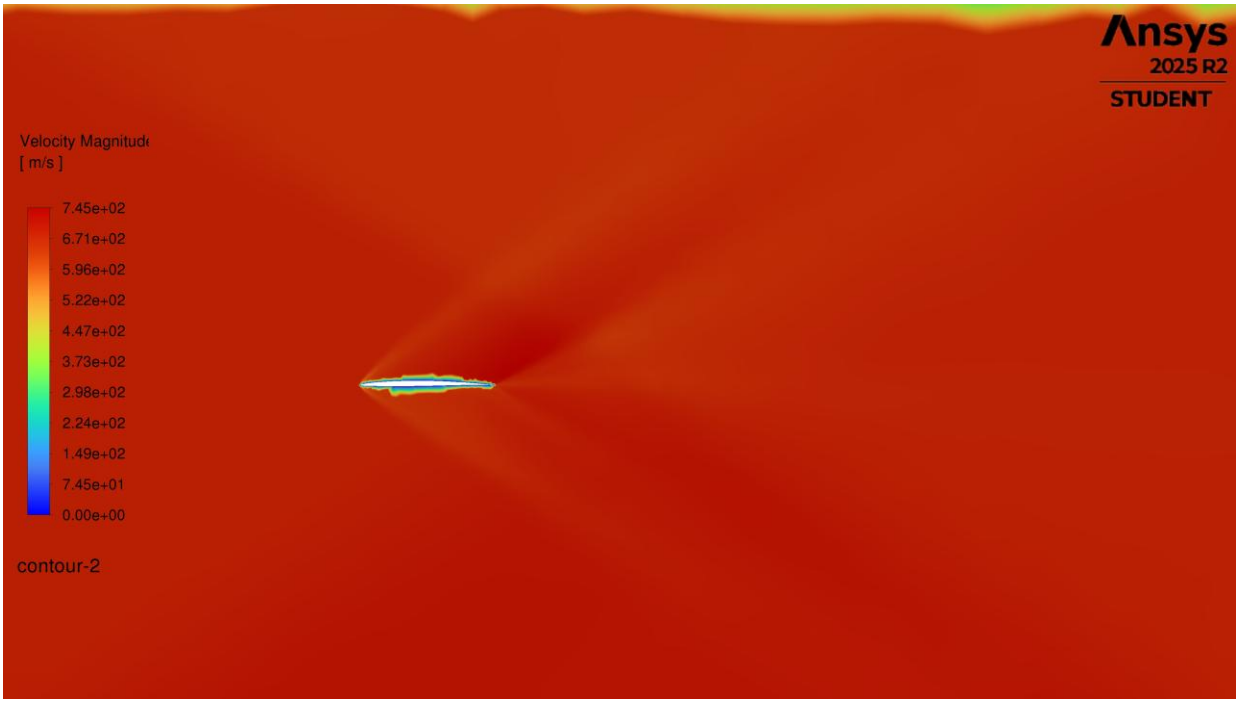
- For pressure at Mach 2 we have max pressure of around 1.29e5 Pa at the leading edge. Overall, we can see that there is lower pressure on the upper edge of the airfoil.



- On our temperature contour we can see that our leading edge once again has the max temperature which looks to be around 3.71e2 Kelvin



- From our velocity contour we can see that the velocity around our airfoil is more or less uniform. It looks to be around 7.45e2 m/s



Conclusion

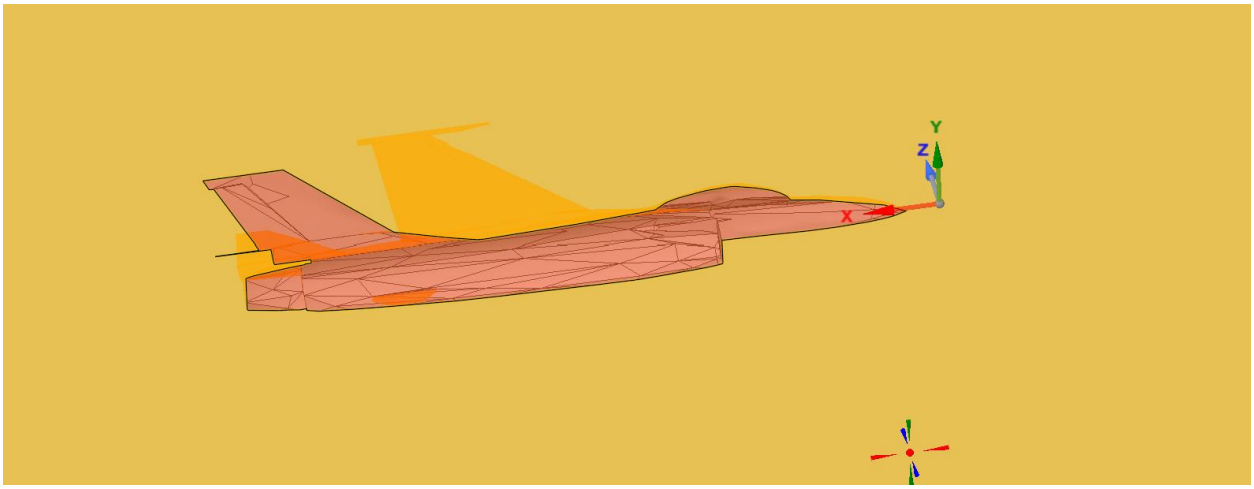
During this section of the project, we analyzed the airfoil section of an F-16 fighter jet. Using Ansys, we were able to calculate the drag and lift force as well as perform an analysis on the surrounding air pressure, temperature, and velocity magnitude under 3 different speeds: cruising, Mach 1, and Mach 2. Although we faced initial difficulties setting up the simulation, after a series of trials and errors, we managed to achieve our results.

F-16 Results

Methodology

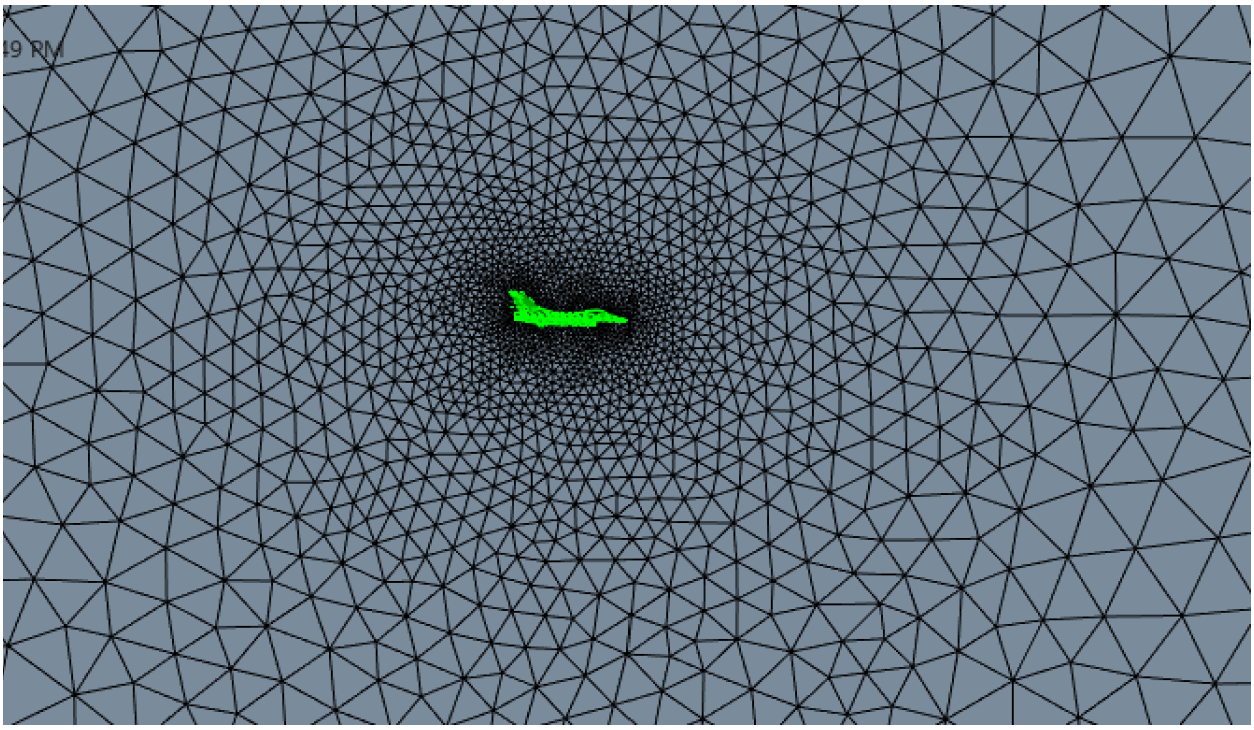
For our aircraft, the chosen model is the F-16 Fighting Falcon, a single-engine, supersonic multirole fighter designed for high maneuverability and performance. The aircraft has a wingspan of 9.96 meters, an average wing chord of approximately 1.3 meters, and a length of 15.06 meters. The F-16 features a blended wing-body design with a cropped delta wing and a leading-edge root extension to improve lift at high angles of attack. We imported the geometry into Ansys and scaled the model to a smaller size in order to be able to get realistic experimental data in a wing tunnel. For our computational setup, we included an enclosure around the aircraft, using the Boolean subtraction function to define the flow domain, which facilitates aerodynamic calculations such as lift, drag, and pressure distribution. For, my geometry i chose the chord

length of the model to be 0.07m which is approximately 2.75 inches. This will allow me to get experimental data that can match the simulations.



To obtain the mesh, we selected the enclosure and made some edits to the regular mesh. First, we made the element size 0.25m, the growth rate 1.1, and the max size 0.25m. Next, we inserted a face sizing and selected the faces of the body, turning “capture curvature” on in the process. This refined the mesh around the wing area so the results will be more accurate.

Variable	Selection
Element type	Default(unchanged)
Element size	0.25m
Number of Elements	144402
Mesh Type	3D
Growth rate	1.1



Setting up the problem was probably the trickiest portion of the CFD. After various trials and errors, we managed to find a working formula. First, we use double precision along with four solver processes as the base. First off, we start off with a density-based solver with steady time. In this CFD we had energy and the viscous (SST K-omega) models turned on. Another change made to the set-up was using ideal-gas properties on the air, with Sutherland's three coefficient equation for viscosity. The boundary condition for the inlet is “velocity-inlet” where we set the components of velocity to those of an aircraft with an angle of attack of 4 degrees. The outlet is set to “pressure-outlet". Initialization used is standard initialization computed from the inlet. It is then run for 300 iterations for accurate and speedy data.

Results

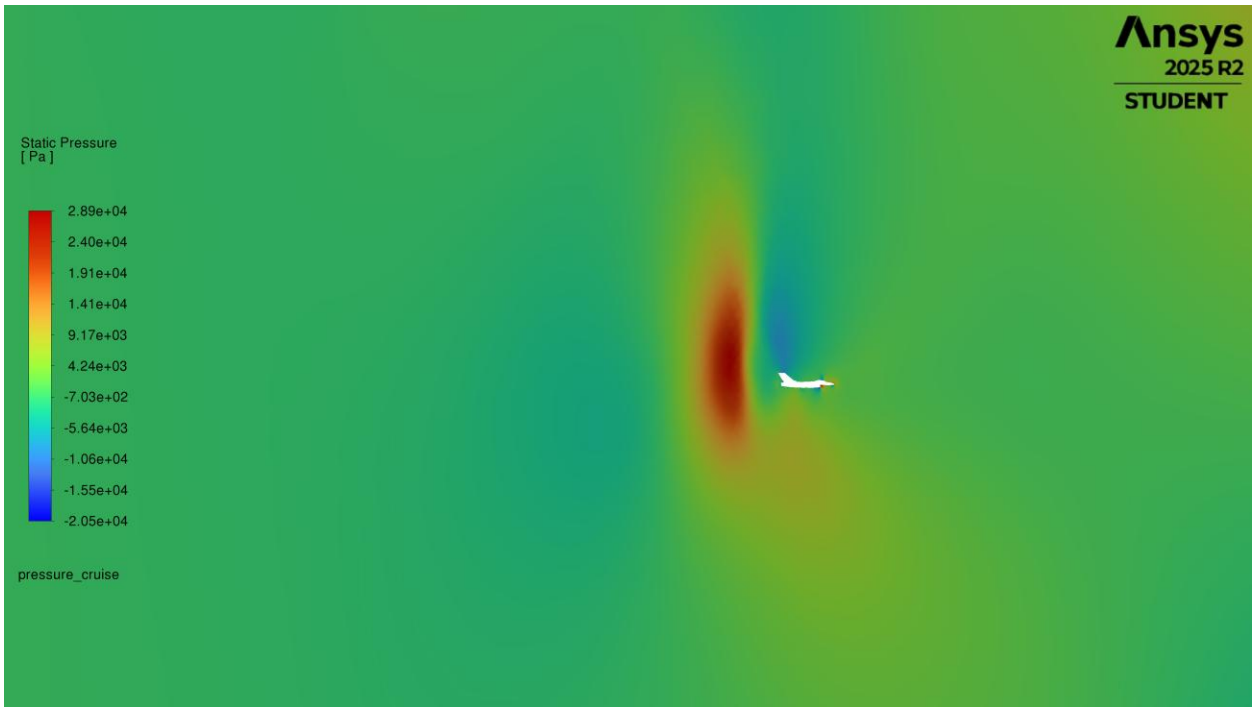
Simulation

Using both post and fluent we were able to obtain good results. As previously mentioned, we analyzed the pressure, temperature and velocity field, as well as calculated the lift and drag using an Ansys' built-in calculator. For this second part, just like the first part, we considered three scenarios. These scenarios are cruising speed, Mach 1, and Mach 2 speeds. Through some research we were able to figure out the cruising speed to be around 275 m/s, and Mach 1 and 2 are 342 m/s and 680 m/s respectively.

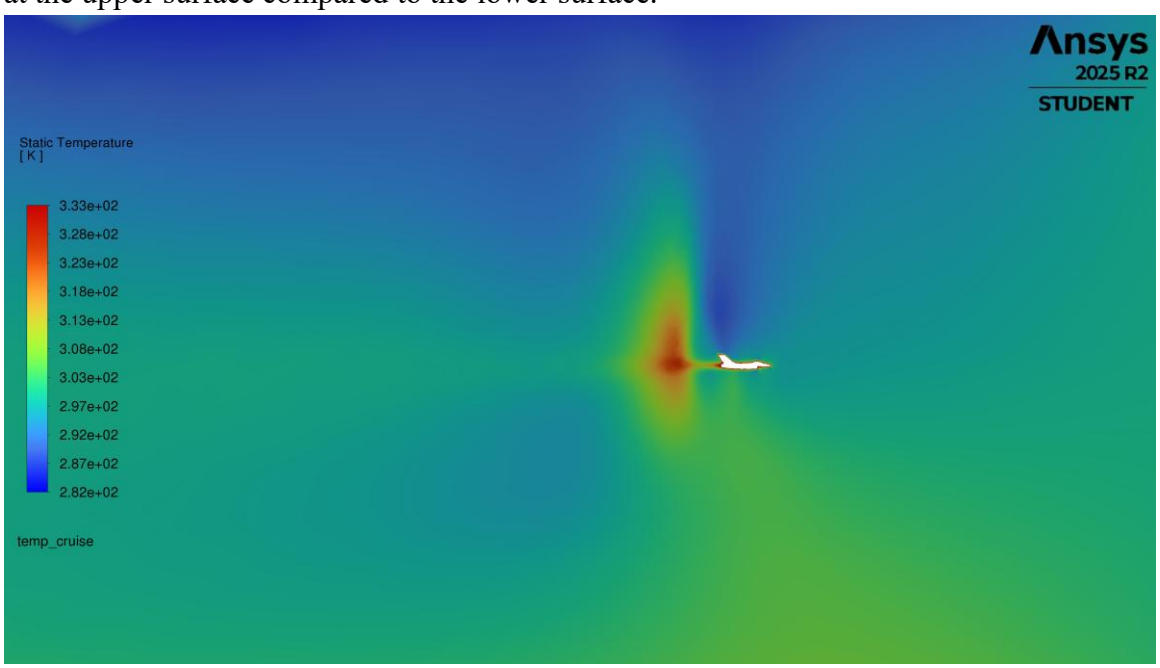
Cruising Speed:

First, we will discuss the results from cruising speed. Here we get a drag force of 7.62502 [N] and a lift force of 41.72[N]. This gives us an L/D ratio of 0.55. The following are the contours:

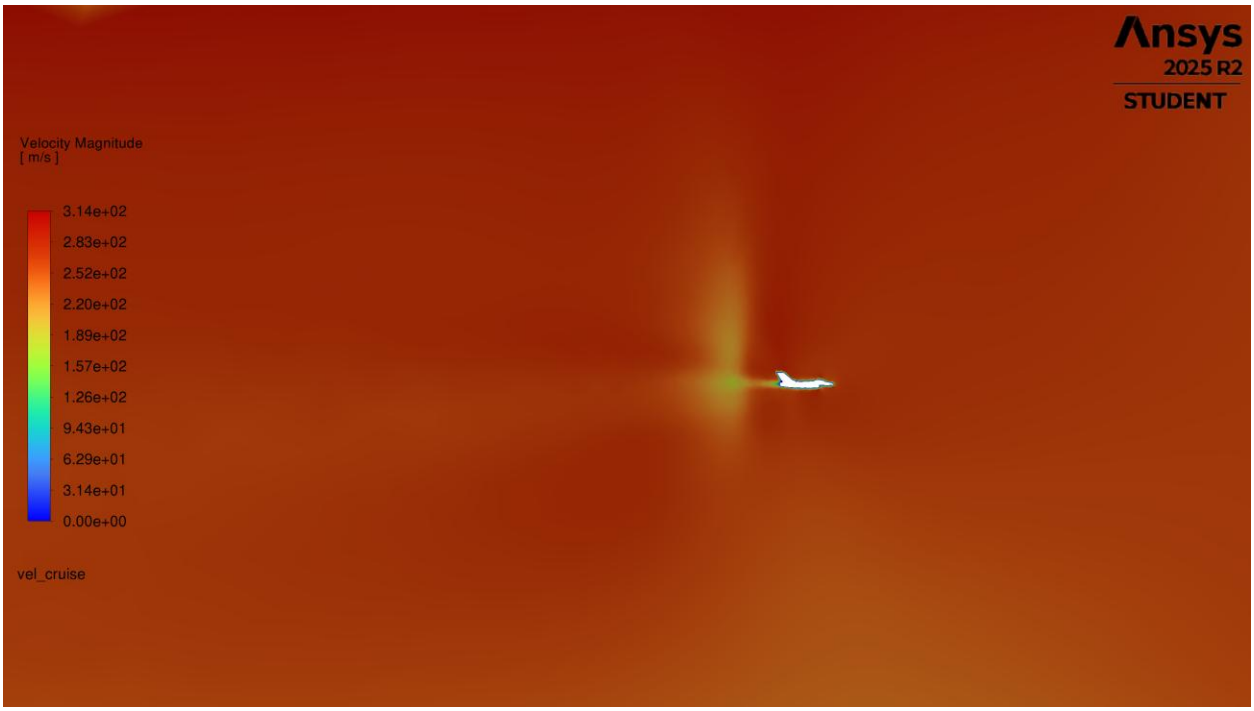
- This would be the pressure contour at a cruising speed. As we can see, we have low pressure area at the upper surface and higher pressure in the lower surface which matches what we expected

- 

- This case is the cruise temperature. Like the pressure, it seems to have a lower value at the upper surface compared to the lower surface.



- This is the velocity contour, here we can see how the speeds around the body look.

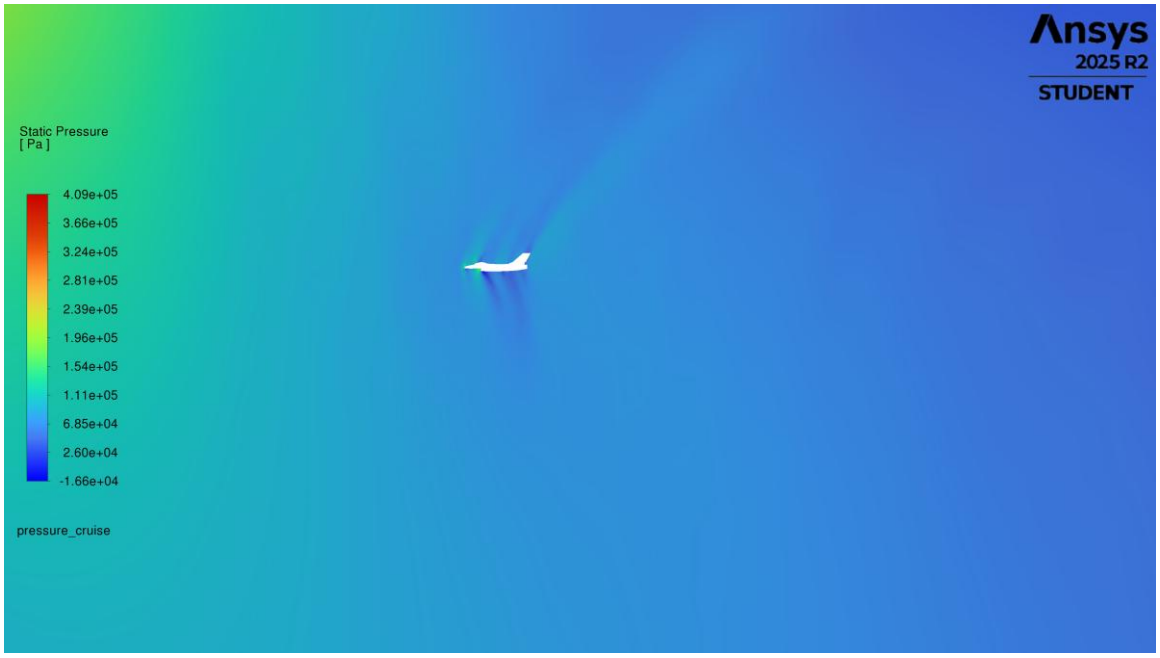


Mach 1:

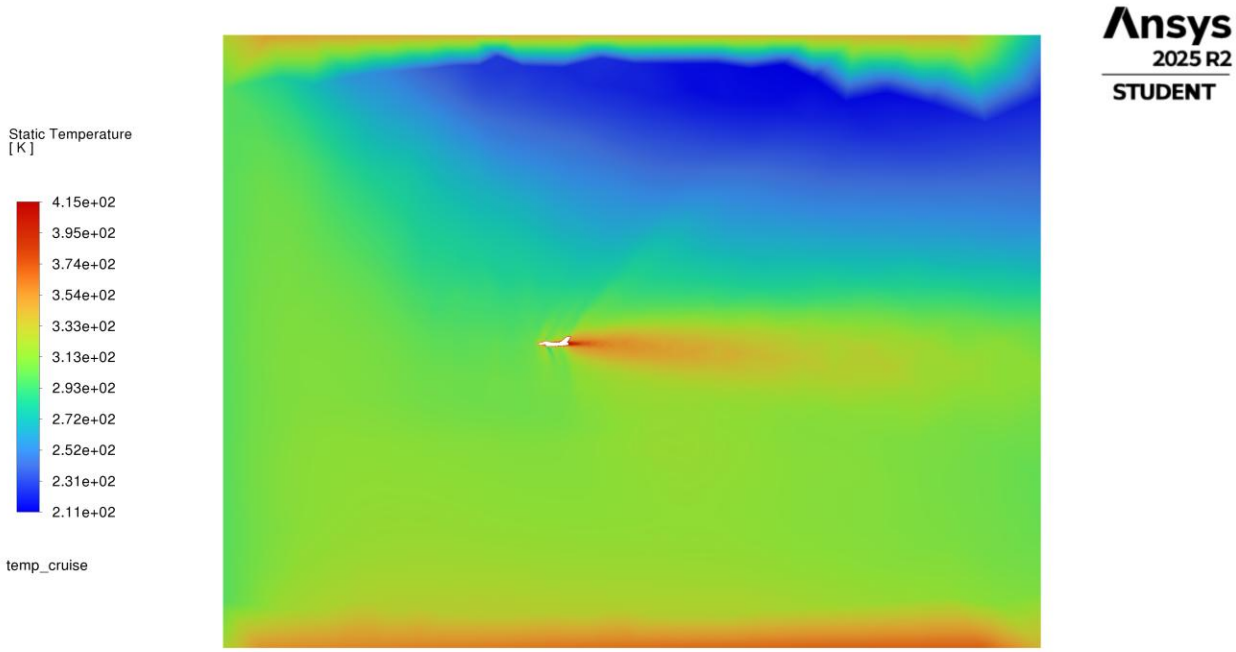
At Mach 1 we achieved a drag force of 3.25[N] and a lifting force of 20.5[N]. This gives us an L/D ratio of 6.31

9. Below will include the pressure, temperature, and velocity contours at this respective speed.

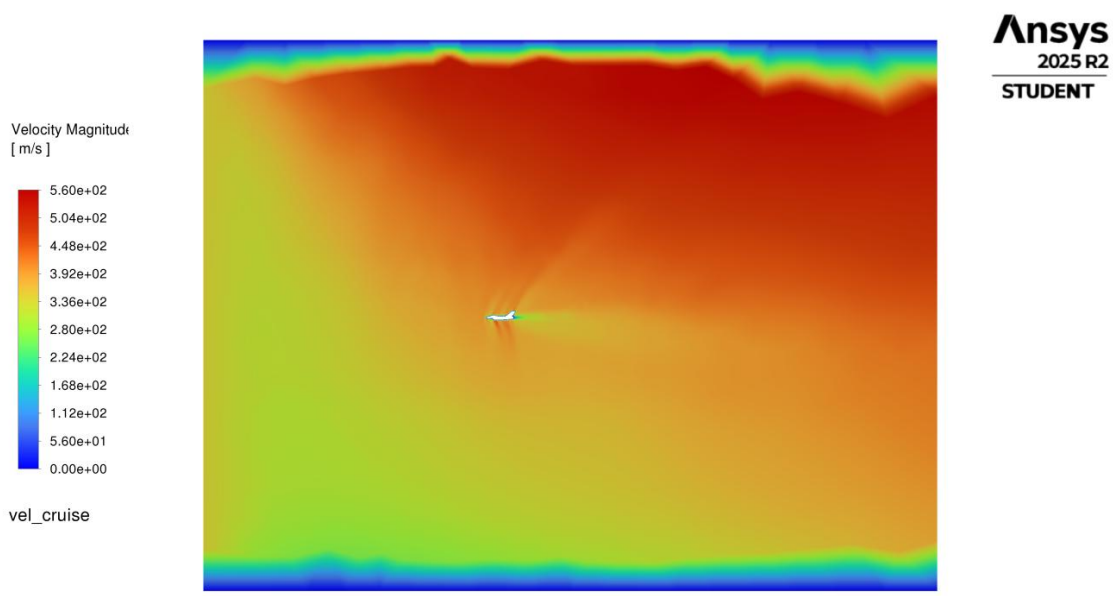
- For static pressure we got a max pressure of around 1.1e5 Pa happening at the leading edge.



- For temperature we have a max temperature happening in the trailing edge of around maybe 4.15e2 kelvin, and a slightly lower temperature of around 3.13e2 Kelvin at the leading edge



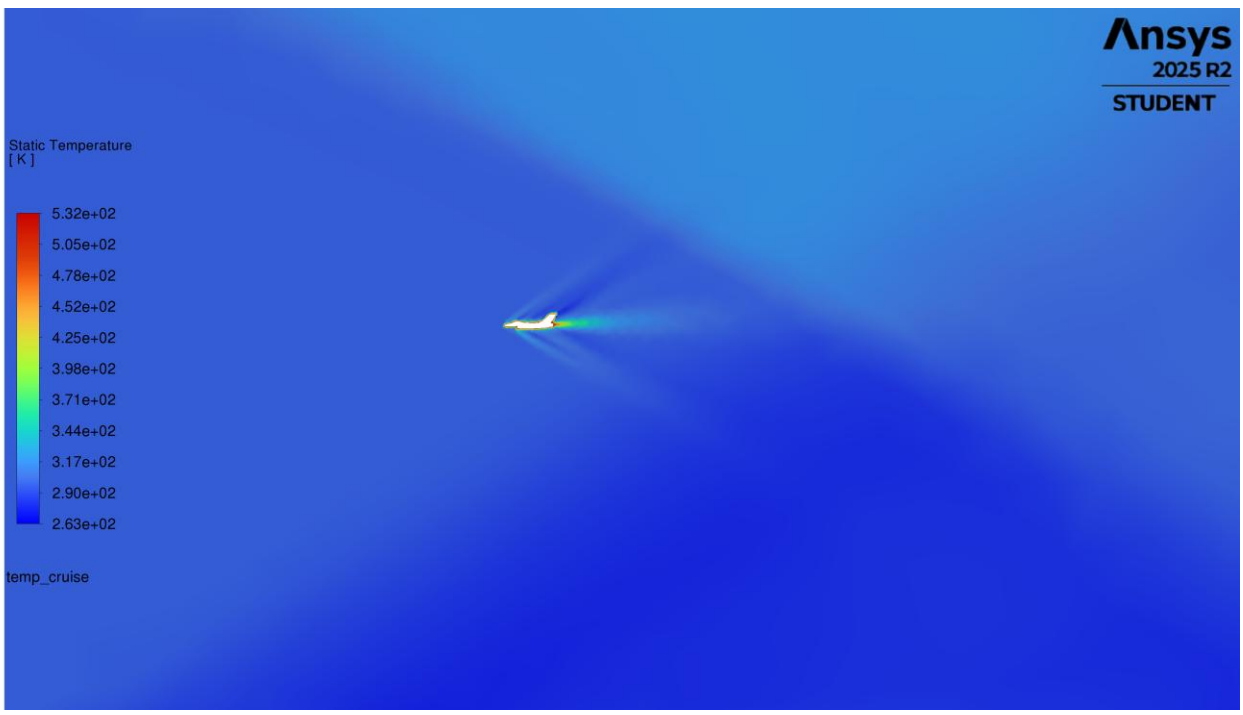
- In the velocity as well as other contours we can see how the shockwaves are starting to form as it breaks the sound barrier.



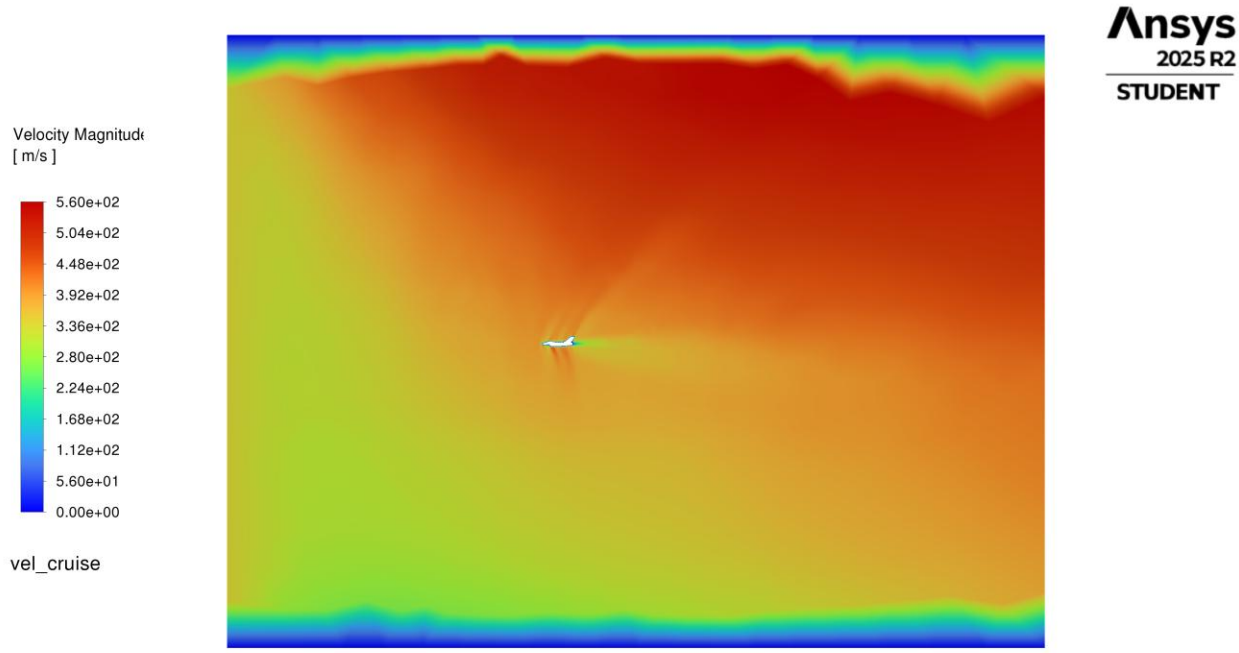
Mach 2:

At Mach 2, calculations gave us a lift of 20.3[N] and a drag of 6.93 [N]. Overall, this gave us an L/D ratio of 2.93. This being lower than that of Mach 1 which matches the airfoil results. Below are the contours for pressure, temperature, and velocity.

- For pressure at Mach 2 we have max pressure of around 1.29e5 Pa at the leading edge. Overall, we can see that there is lower pressure on the upper edge of the airfoil.



- On our temperature contour we can see that our leading edge once again has the max temperature which looks to be around 3.71e2 Kelvin
- From our velocity contour we can see that the velocity around our airfoil is more or less uniform. It looks to be around 7.45e2 m/s



Experimental

In order to get experimental data, we printed a 0.07m or about 2.75” model of our F16. Using the Lockheed Martin wind tunnel, we were able to run 3 different trials on our model. The first trial the velocity was set to 11m/s. This gave us a lift of 1.3[N] and a drag of 0.1[N] offering and L/D ratio 13. The second trial set the speed to 14 m/s this gave us a resultant lift of 1[N] and drag of 0.1[N], this gives us an L/D ratio of 10. The last trial’s wind velocity was set to 16 m/s. We got a lift of 0.8[N] and drag 0.2[N], overall giving an L/D ratio of 4. These results make sense, since the model was very small in comparison to the actual size and the wind speed was also very low. At a top speed of 16m/s and a very small surface are of the model, it makes sense as to why both the generated lift and drag would be so small.

Conclusion

Throughout the project, we were able to apply what we learned in our aerodynamics lectures to real-world scenarios. Both the experimental and simulated data helped solidify and strengthen our understanding of aerodynamic principles. Seeing how concepts like boundary layers, pressure gradients, and drag coefficients played out in practice reinforced the material covered in class. Despite the limitations of our setup, these results

helped us better understand how theoretical models compare to practical outcomes. We observed clear relationships between velocity and drag, pressure, and temperature in the simulations, and through the experiments we were able to collect real-world data, even if it was at relatively low velocities. This project also improved our ability to interpret flow behavior, analyze trends, and troubleshoot issues in both simulation and testing environments. Overall, the project gave us a much clearer picture of how aerodynamic theory translates into measurable physical behavior.